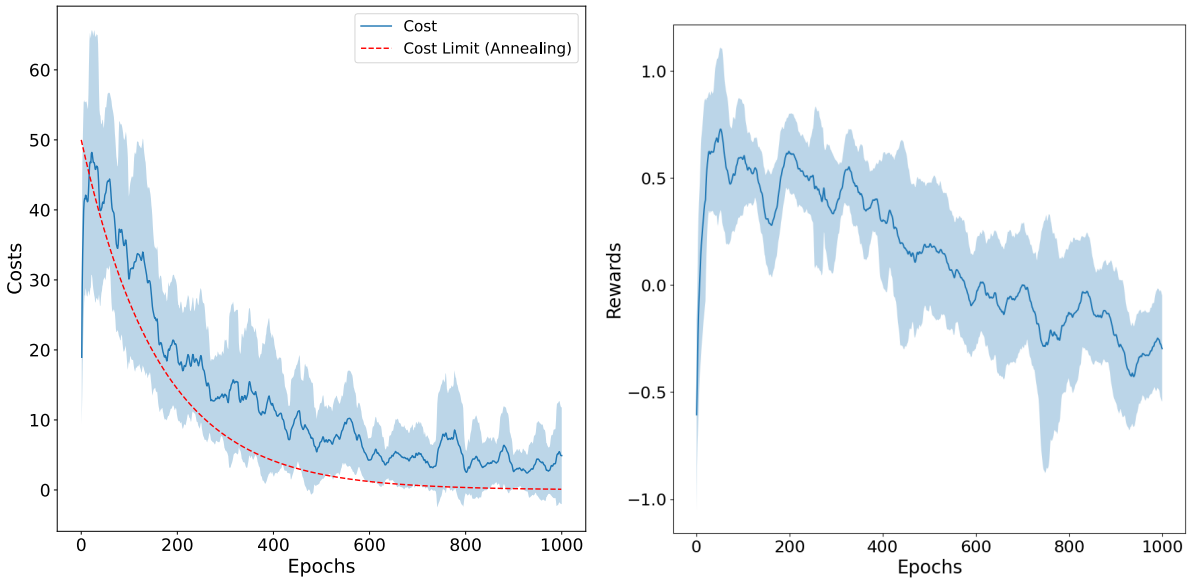
Table 8. Selected baselines using both a fixed relaxed cost limit of 15 and annealing schedules (exponential decay from 50 to 0.1 and piecewise linear: first third episodes at 15, last third at 0.1, middle third decreasing linearly). Red: negative (safe) reward, italics: safety probability $< 50\%$, grey: Pareto-dominated pairs, bold: best value per metric (only among methods with positive (safe) reward and safety probability $\geq 50\%$ for the other metric).

Method	Cost limit	Point Push		Point Goal		Car Circle		Swimmer Velocity	
		Safe Rew.	Safe Prob.	Safe Rew.	Safe Prob.	Safe Rew.	Safe Prob.	Safe Rew.	Safe Prob.
Ours	–	0.33 ± 0.18	$\mathbf{0.79 \pm 0.08}$	$\mathbf{1.6 \pm 0.5}$	0.68 ± 0.07	7.5 ± 1.5	$\mathbf{0.99 \pm 0.01}$	48 ± 26	$\mathbf{0.98 \pm 0.02}$
TRPO-Lag	0	-0.49 ± 0.45	0.78 ± 0.16	0.09 ± 0.29	$\mathbf{0.79 \pm 0.06}$	11 ± 4	0.81 ± 0.26	$\mathit{1.3 \pm 5.1}$	$\mathit{0.33 \pm 0.42}$
TRPO-Lag	exp	-0.55 ± 0.47	0.84 ± 0.11	0.54 ± 0.79	0.77 ± 0.072	$\mathbf{15 \pm 0.15}$	0.88 ± 0.084	$\mathit{-4.6 \pm 9.3}$	$\mathit{0.19 \pm 0.38}$
TRPO-Lag	lin	-0.24 ± 0.83	0.83 ± 0.087	0.5 ± 1.1	0.76 ± 0.078	14 ± 1.5	0.91 ± 0.13	11 ± 21	0.51 ± 0.36
TRPO-Lag	15	0.29 ± 0.28	0.64 ± 0.12	-0.18 ± 0.53	0.82 ± 0.08	$\mathit{6.8 \pm 1.5}$	$\mathit{0.44 \pm 0.09}$	$\mathit{0.0 \pm 0.0}$	$\mathit{0.0 \pm 0.0}$
CPO	0	-1.2 ± 0.9	0.77 ± 0.19	-1.3 ± 0.4	0.86 ± 0.05	0.96 ± 0.76	$\mathbf{0.99 \pm 0.03}$	-3.8 ± 9.3	0.99 ± 0.03
CPO	exp	-0.35 ± 0.26	0.87 ± 0.047	-0.85 ± 0.59	0.87 ± 0.05	14 ± 0.49	0.97 ± 0.031	$\mathbf{69 \pm 62}$	0.72 ± 0.36
CPO	lin	-0.7 ± 0.29	0.89 ± 0.054	-0.84 ± 0.45	0.84 ± 0.062	12 ± 3.2	0.97 ± 0.028	28 ± 1.7	0.89 ± 0.059
CPO	15	0.24 ± 0.14	0.61 ± 0.07	0.29 ± 0.67	0.69 ± 0.11	8.3 ± 1.6	0.54 ± 0.11	$\mathit{0.0014 \pm 0.0098}$	$\mathit{0.001 \pm 0.0052}$
C-TRPO	0	-1.4 ± 1.2	0.83 ± 0.09	-2.2 ± 1.1	0.88 ± 0.06	0.54 ± 0.5	0.97 ± 0.05	-18 ± 8	0.97 ± 0.03
C-TRPO	exp	-0.61 ± 0.32	0.88 ± 0.042	-1.6 ± 0.84	0.83 ± 0.071	13 ± 0.59	0.95 ± 0.063	52 ± 49	0.90 ± 0.06
C-TRPO	lin	-0.55 ± 0.59	0.86 ± 0.083	-1.2 ± 0.41	0.84 ± 0.065	14 ± 0.61	0.96 ± 0.036	28 ± 14	0.73 ± 0.37
C-TRPO	15	$\mathbf{0.48 \pm 0.3}$	0.58 ± 0.08	1.2 ± 0.5	0.66 ± 0.08	11 ± 1	0.69 ± 0.08	$\mathit{0.05 \pm 0.16}$	$\mathit{0.0092 \pm 0.0223}$

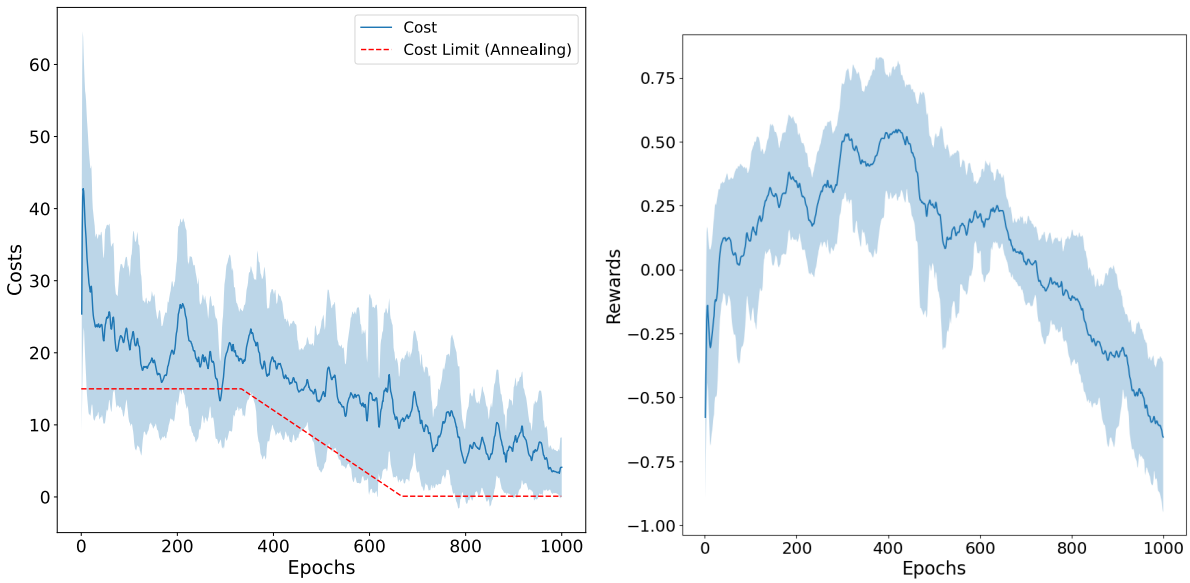
Table 9. Effect of using critics on performance. For each method, reward and cost are shown with and without a critic; the worse-performing variant per method is coloured in grey.

Method	Critic?	Point Button		Car Goal		Point Goal	
		Reward$^{\uparrow}$	Cost$^{\downarrow}$	Reward$^{\uparrow}$	Cost$^{\downarrow}$	Reward$^{\uparrow}$	Cost$^{\downarrow}$
Ours	no	1.9	15	1.1	16	2.4	18
Ours	yes	5.0	36	5.9	38	5.0	25
TRPO-Lag	no	0.39	16	0.49	24	0.77	30
TRPO-Lag	yes	-0.72	15	0.29	16	0.11	17
CPO	no	-3.2	5.8	-0.89	13	-1.8	11
CPO	yes	-3.5	6.1	-0.92	9.2	-1.5	9.3
C-TRPO	no	-2.3	6	-2.1	8.5	-2.2	17
C-TRPO	yes	-3.0	9.4	-1.0	6.9	-2.5	8.7



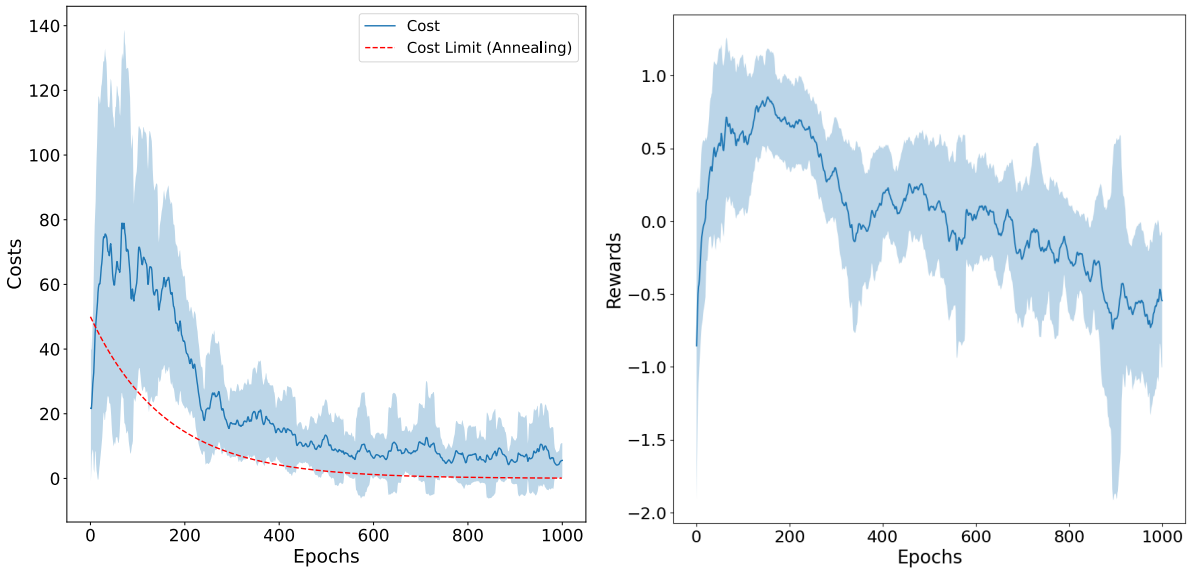
(a) Cost *(b)* Reward

Figure 11. Training curves for CPO with exponential annealing



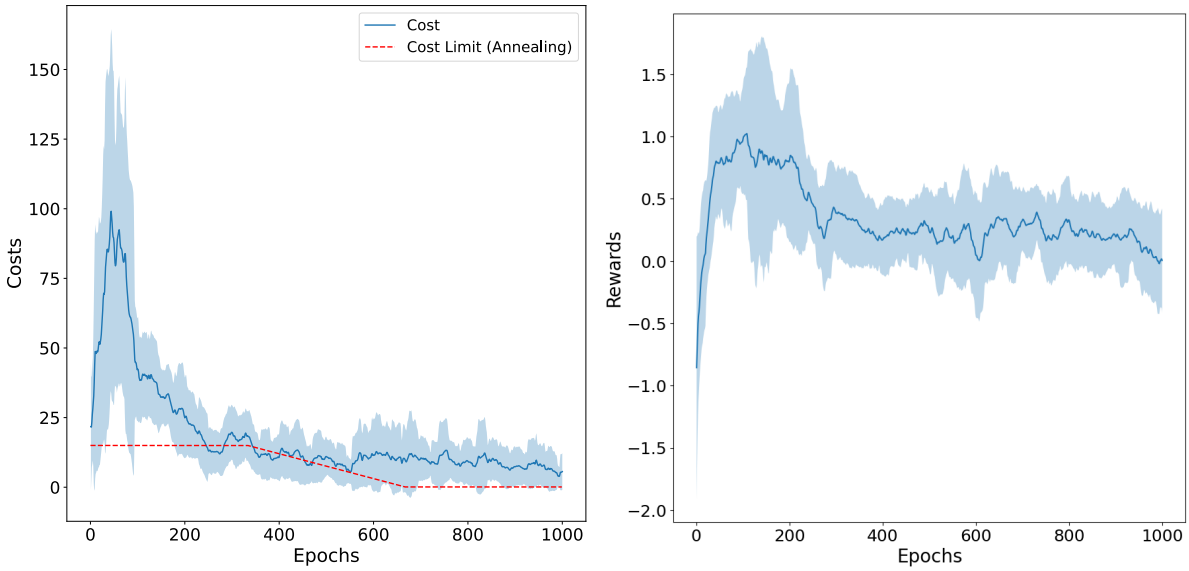
(a) Cost *(b)* Reward

Figure 12. Training curves for CPO with piecewise linear annealing



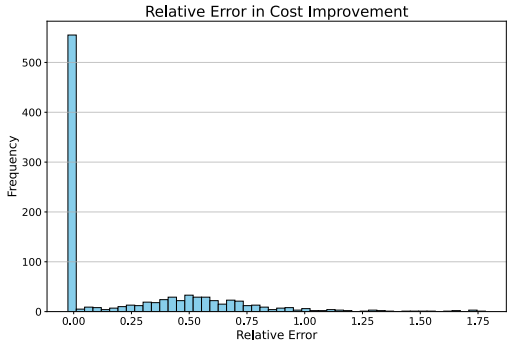
(a) Cost *(b)* Reward

Figure 13. Training curves for TRPO-Lag with exponential annealing

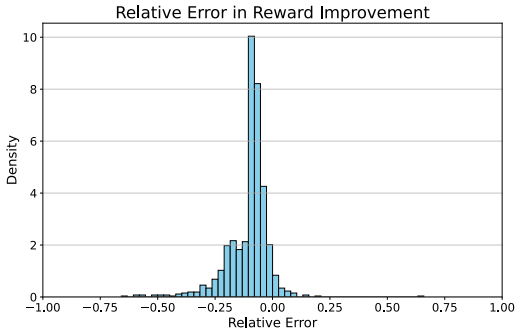


(a) Cost *(b)* Reward

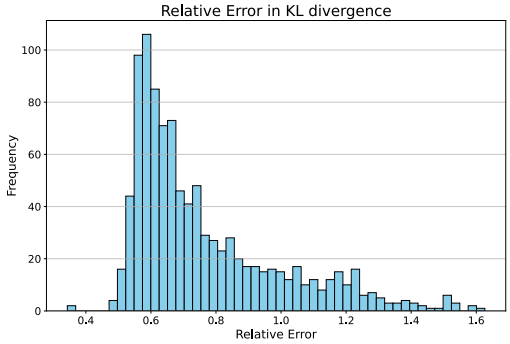
Figure 14. Training curves for TRPO-Lag with piecewise linear annealing



(a) Cost



(b) Reward



(c) KL divergence

Figure 15. Histograms of relative approximation errors ($\frac{\widehat{x}-x}{x}$).